

ADZE: Allelic Diversity Analyzer

Version 2.0

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Abstract

ADZE implements a rarefaction approach to estimating allelic richness, private allelic richness, and private allelic richness for combinations of populations, correcting for differences in sample size between populations. Version 2.0 computes the same estimators as version 1.0 — results are identical to the last bit — but reads its data in one pass, caches the rarefaction probabilities, parallelizes over loci, and replaces the parameter file with a conventional command-line interface. Section 8 lists everything that changed.

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1 Introduction

The analysis of the distributions of alleles across populations is important for elucidating genetic diversity and population relationships. Two fundamental quantities for a population at a given locus are the number of distinct alleles in the population and the number of alleles found in that population and no other. Both are strongly dependent on sample size: a larger sample of gene copies is more likely to contain any given allele, so counts taken from samples of different sizes cannot be compared directly.

Rarefaction addresses this by computing, for a standardized sample size g , the expected number of alleles that *would* be observed if g gene copies were drawn from each population. The estimators ADZE computes are exact expectations under sampling without replacement, not resampling averages, so no random number generator and no seed are involved.

ADZE implements these computations for allelic richness, private allelic richness, and a generalization of private allelic richness to combinations of populations, as described in Szpiech et al. (2008).

1.1 Allelic richness

Consider a locus with I distinct alleles, and define N_{ij} as the number of copies of allele type i in a sample from population j . Let $N_j = \sum_{i=1}^I N_{ij}$ be the sample size of population j at the locus. The probability of finding no copies of allele type i in a subsample of g gene copies from population j is

$$Q_{ijg} = \frac{\binom{N_j - N_{ij}}{g}}{\binom{N_j}{g}}. \quad (1)$$

The probability of finding at least one copy of allele type i in a sample of size g from population j is $P_{ijg} = 1 - Q_{ijg}$, and

$$\hat{\alpha}_g^{(j)} = \sum_{i=1}^I P_{ijg} \quad (2)$$

is the estimated allelic richness of a sample of size g from population j (Hurlbert, 1971; Petit et al., 1998; Kalinowski, 2004). Equation 2 estimates the expected number of distinct alleles that will be observed in population j in a sample of size g .

1.2 Private allelic richness

Using this notation, the estimated private allelic richness for a sample of size g from population j is

$$\hat{\pi}_g^{(j)} = \sum_{i=1}^I \left[P_{ijg} \left(\prod_{\substack{j'=1 \\ j' \neq j}}^J Q_{ij'g} \right) \right], \quad (3)$$

where J is the total number of populations (Kalinowski, 2004). This formula sums over distinct allele types i the probability that a random subsample of size g from population j contains allele type i and that subsamples of size g from the remaining populations do not.

1.3 Private allelic richness for a combination of populations

Generalizing the concept of private allelic richness, we can consider the number of distinct alleles private to some combination of k populations selected from $\{1, 2, \dots, J\}$. Let $S = \{1, 2, \dots, J\}$ and let C_k be the set of all combinations of k elements from S ; there are $\binom{J}{k}$ of them. Labelling these combinations C_{km} , where m ranges from 1 to $\binom{J}{k}$, the estimated number of distinct alleles private to the m th combination of k populations, when samples of size g are drawn from each of the J populations, is

$$\hat{\pi}_{gk}^{(m)} = \sum_{i=1}^I \left[\left(\prod_{j \in C_{km}} P_{ijg} \right) \left(\prod_{j' \in S \setminus C_{km}} Q_{ij'g} \right) \right]. \quad (4)$$

Here $S \setminus C_{km}$ denotes the set S excluding the elements of C_{km} . For $k = 1$, $\hat{\pi}_{gk}^{(m)}$ reduces to private allelic richness as in Equation 3. For $k = J - 1$, Equation 4 can be considered a measure of “missing allelic richness”, and it reduces to

$$\hat{\mu}_g^{(j)} = \sum_{i=1}^I \left[Q_{ijg} \left(\prod_{\substack{j'=1 \\ j' \neq j}}^J P_{ij'g} \right) \right], \quad (5)$$

the expected number of alleles absent from population j but present in every other population.

1.4 Summary statistics

ADZE evaluates Equations 2–4 at each of many loci in a data set, and its primary method of reporting results gives the mean, variance, and standard error of the mean across these loci. Locus-specific estimates can also be written out (section 5.3). The mean across loci is calculated at each g and for each population or population grouping as

$$\bar{x} = \frac{1}{L} \sum_{\ell=1}^L x_{\ell}, \quad (6)$$

where L is the number of loci used in the calculation and x_{ℓ} is one of the statistics above evaluated at locus ℓ . The variance across loci is

$$s^2 = \frac{1}{L-1} \sum_{\ell=1}^L (x_{\ell} - \bar{x})^2, \quad (7)$$

and the standard error of the mean is

$$\text{SE} = \frac{s}{\sqrt{L}}. \quad (8)$$

A statistic is reported at a given g only if it is defined at that g for every locus used, which requires $g \leq N_j$ at each locus for the grouping concerned. With a single locus ($L = 1$) the variance and standard error are undefined and are reported as **nan**.

2 Getting started

2.1 Obtaining adze

Source is at <https://github.com/szpiech/ADZE>. Version 2.0 requires nothing but a C++17 compiler. Two libraries are used when the toolchain provides them, and the program builds and

runs without either: OpenMP for `--threads`, and zlib for compressed input. Without zlib a `.gz` file is refused with a message rather than misread.

2.2 Building

With CMake:

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build -j
cmake --install build --prefix ~/.local      # optional
```

Or with plain make:

```
make -C src                # serial, no compressed input
make -C src OPENMP=1 ZLIB=1 # GCC, or clang on Linux
make -C src OPENMP=1 ZLIB=1 OMPFLAGS="-Xpreprocessor -fopenmp" OMPLIBS="--lomp"
                           # Apple clang, with libomp installed
```

CMake reports at configure time which of the two it found; the `make` build takes each as a flag. Without OpenMP, `--threads` is accepted but warns that it is being ignored.

2.3 Running

There are two ways to specify a run, and they can be mixed:

```
adze --data mydata.stru --group-col 5 --out-prefix mypops
adze myparamfile.txt --tolerance 0.05
```

The first form takes everything from the command line. The second reads a parameter file in the version 1.0 format — the same keywords, and the same short flags — with command-line options taking precedence over the file. Three sources are consulted, in increasing order of precedence: built-in defaults, then the parameter file if one is given, then the command line.

`--help` lists every option; `--version` prints the version and the method reference. Both work with no data file and no parameter file. `--write-template` writes a commented parameter file that runs as it stands once `DATA.FILE` is filled in.

Only the data file is strictly required. Everything else either has a default or is measured from the data (section 3.3).

3 Input

3.1 Choosing a format

Two input formats are read: the STRUCTURE-like layout that ADZE has always taken (section 3.2), and VCF (section 3.4), optionally compressed with `gzip` or `bgzip`. `--format` selects one; its default, `auto`, reads the file name — anything ending `.vcf`, after an optional `.gz` or `.bgz`, is treated as VCF.

The two paths converge immediately: both reduce each locus to per-grouping allele counts and missing-data tallies, which is all the estimators use, and both lay those counts out identically for the same genotypes. Results therefore do not depend on which format the data arrived in — the same genotypes written both ways produce byte-identical output, which `test/formats.py` checks.

3.2 structure format

This file is expected in the same format used by STRUCTURE: one row per gene copy, so a diploid individual occupies two consecutive rows, preceded by a header row of locus names. Each data row begins with one or more label columns and continues with one allele code per locus. Allele codes are arbitrary tokens — integers, as in the example below, or any other strings that do not contain whitespace.

Figure 1 shows `example/small_data.stru`, distributed with ADZE: five loci, three groupings, and eleven diploid individuals, with missing alleles coded -9.

```
D12S1638 D14S1007 D9S1779 D9S1825 D7S2477
1037 87 Pima Mexico AMERICA 126 128 124 -9 142
1037 87 Pima Mexico AMERICA 126 128 142 -9 142
1038 87 Pima Mexico AMERICA 120 124 124 129 142
1038 87 Pima Mexico AMERICA 126 128 152 137 142
...
1357 24 Basque France EUROPE 126 120 124 -9 154
1357 24 Basque France EUROPE 128 122 126 -9 156
...
747 684 Japanese Japan EAST_ASIA 126 -9 124 129 158
747 684 Japanese Japan EAST_ASIA 128 -9 144 135 158
...
```

Figure 1: The distributed example data file, abridged. Row 1 holds the locus names; each subsequent row is one gene copy, with five label columns followed by five allele codes.

3.3 Dimensions are measured, not declared

Version 1.0 required `LOCI`, `DATA_LINES`, `NON_DATA_ROWS` and `NON_DATA_COLS` to be supplied by hand, and validated them against the file. Version 2.0 measures them while reading:

`--loci`

from the number of names in the header row.

`--data-lines`

by counting data rows.

`--non-data-cols`

from the width of the first data row minus the locus count.

`--group-col`

defaults to the last label column.

`--non-data-rows`

defaults to 1, the locus-name row. This is the one value that cannot be inferred, so give it explicitly if the file has additional header rows.

`--max-g`

defaults to the largest standardized sample size the data supports, which is the smallest N_j over all groupings and all *surviving* loci — resolved after missing-data filtering, since dropping loci can raise it.

Any of these may still be given, in which case the declared value is used and a warning is printed if it disagrees with what was measured. `--dry-run` reports the detected layout, the per-grouping sample sizes, the largest feasible `MAX_G` and the number of tuples an analysis would evaluate, then exits without computing anything.

3.4 VCF format

Each record is one locus and each allele index is one allele type, so the reference allele is allele 0 and the alternates follow. A sample contributes as many gene copies as its `GT` field holds, which lets haploid and diploid records sit in the same file, and a `.` is an uncalled copy. A sample's ploidy is the largest number of alleles any of its `GT` fields holds, since that is how many gene copies it could have contributed; a record where it carries fewer — a haploid call in an otherwise diploid file, say — counts the difference as missing, and so tells against that locus under `--tolerance`. Only `GT` is read: no other `FORMAT` field, and no `INFO`, `QUAL` or `FILTER` value, affects the counts, so filtering must be done before ADZE sees the file. Locus names come from the `ID` column, or from `CHROM:POS` where `ID` is `..`.

A VCF names its samples but says nothing about which population each belongs to, so `--samples` is required with VCF input. It takes a file of two whitespace-separated columns, sample name then grouping name:

```
# sample      grouping
HG00096      GBR
HG00097      GBR
NA18939      JPT
```

Anything after `#` is a comment and any further columns are ignored, so an existing sample sheet can usually be passed as it stands. Groupings are numbered in the order they first appear in this file, which is the order they appear in the output. A sample in the VCF that the map does not mention is skipped, and a sample the map names that the VCF does not contain is ignored; both are reported as warnings with a count.

Because a sample's gene copies are what the missing-data fraction is measured against, `--tolerance` means the same thing here as for `STRUCTURE` input: the fraction of a grouping's gene copies that are not called at that locus (section 3.6).

```
adze --data cohort.vcf.gz --samples populations.tsv \
    --tolerance 0.05 --out-prefix cohort
```

`NON_DATA_ROWS`, `NON_DATA_COLS` and `GROUP_BY_COL` describe the `STRUCTURE` layout and are ignored, with a warning, for VCF input.

3.5 Groupings

With `STRUCTURE` input, rows are grouped by the value in the `--group-col` label column; with VCF input the grouping comes from `--samples`. In both cases groupings appear in output in the order they are first seen; any label column can be used, so a file annotated with population, region and continent can be analysed at any of those levels without being rewritten, and the same VCF can be analysed at several levels by swapping the map file. `--pops` restricts the analysis to a comma-separated list of groupings and `--exclude-pops` removes them; both change J , and therefore change what “private” means, so they are a way of asking a different question rather than a way of subsetting output.

3.6 Missing data and locus filtering

An allele code equal to `--missing` (default `-9`) is treated as unobserved and reduces N_j at that locus. Because a statistic is reported at a given g only when it is defined at every locus, one locus with a small N_j limits the whole analysis.

`--tolerance  $x$` drops any locus at which some grouping has more than a fraction x of its gene copies missing. The default, $x = 1$, keeps every locus. Loci dropped this way are listed in the `_deletedloci` file (section 5.4). Filtering is a trade: it discards loci to buy a larger `MAX_G`, and section 6.1 works through the trade-off on the example data.

A locus at which no allele is observed in any grouping contributes nothing; such loci are counted, reported in a warning, and treated as undefined at every g . If filtering removes every locus, ADZE stops with an error rather than reporting statistics over no data.

4 Parameters and options

4.1 Parameter file syntax

A parameter file is a sequence of lines, each a keyword followed by its value:

```
DATA_FILE small_data.stru
GROUP_BY_COL 5
MAX_G 8
TOLERANCE 0.24
# everything after a hash is a comment
```

Anything after `#` is ignored. The value is the rest of the line with surrounding whitespace removed, so values may contain spaces and may contain other keywords as substrings. Boolean parameters take 0 or 1.

4.2 Option reference

The following is generated from the option table in the source, so it lists exactly what the program accepts. Each entry gives the long option, the version 1.0 short flag where one exists, and the parameter-file keyword where one exists.

4.2.1 Input options

```
--data FILE, -f, DATA_FILE (paramfile)
    string
    genotype file: STRUCTURE layout or VCF, optionally gzipped (required)

--format FMT, FORMAT (paramfile)
    string
    input format: auto, structure or vcf (default auto, by file extension)

--samples FILE, SAMPLE_FILE (paramfile)
    string
    sample-to-grouping map, required for VCF input (two columns: sample grouping)

--max-g N, -g, MAX_G (paramfile)
    int
    largest standardized sample size (default: the largest the data supports)
```

--non-data-rows *N*, **-nr**, NON_DATA_ROWS (**paramfile**)
int
header rows before the genotypes (default 1)

--non-data-cols *N*, **-nc**, NON_DATA_COLS (**paramfile**)
int
label columns before the genotypes (default: detected)

--loci *N*, **-l**, LOCI (**paramfile**)
int
number of loci (default: detected from the header row)

--data-lines *N*, **-d**, DATA_LINES (**paramfile**)
int
number of data rows (default: detected)

--group-col *N*, **-s**, GROUP_BY_COL (**paramfile**)
int
which label column names the grouping (default: the last one)

--missing *STR*, **-m**, MISSING (**paramfile**)
string
code for a missing allele (default -9)

--pops *LIST*
string
analyse only these groupings (comma-separated)

--exclude-pops *LIST*
string
analyse everything except these groupings

4.2.2 Analysis options

--stat *LIST*
string
which statistics to compute: richness,private,tuples (default: richness,private)

--tolerance *X*, **-t**, TOLERANCE (**paramfile**)
float
drop a locus if any grouping exceeds this missing fraction (default 1 = keep all)

--combinations, **-c**, COMB (**paramfile**)
boolean
also compute private alleles of grouping tuples

--tuples-k *LIST*, **-k**, K_RANGE (**paramfile**)
string
tuple sizes for **-combinations**, e.g. 2 or 1-3 or 1,3,5-7

--tuples *FILE*
string
compute only the named tuples in *FILE* (one per line) instead of every k-subset

--threads *N*
int
worker threads (default 1; requires an OpenMP build)

4.2.3 Output options

--out-prefix *PREFIX*
string
compose output names from PREFIX (default adze)

--out-richness *FILE*, **-r**, *R_OUT* (**paramfile**)
string
allelic richness output file

--out-private *FILE*, **-p**, *P_OUT* (**paramfile**)
string
private allelic richness output file

--out-tuples *FILE*, **-o**, *C_OUT* (**paramfile**)
string
tuple output file (one per tuple size, suffixed *_k*)

--full-richness, **-fr**, *FULL_R* (**paramfile**)
boolean
also write per-locus allelic richness

--full-private, **-fp**, *FULL_P* (**paramfile**)
boolean
also write per-locus private allelic richness

--full-tuples, **-fc**, *FULL_C* (**paramfile**)
boolean
also write per-locus tuple values

--tsv
boolean
tab-separated output with a header row and NA for undefined values

4.2.4 Other options

--params *FILE*
string
read parameters from FILE (also accepted as the first argument)

--dry-run
boolean
report the detected layout, groupings and feasible MAX_G, then stop

--progress, **-pp**, *PRINT_PROGRESS* (**paramfile**)
boolean
progress bars (default: on when stderr is a terminal)

--quiet
boolean
 suppress progress and informational messages

--tolerance-only, -tnocalc, TNC (paramfile)
boolean
 apply the missing-data filter, report it, and stop

--no-check, -skipchk, SKIP_CHK (paramfile)
boolean
 accepted for compatibility; validation is now free and always on

5 Output

By default ADZE writes `adze.richness`, `adze.private` and `adze.summary.txt` in the current directory; **--out-prefix** replaces the `adze` stem. Giving an explicit name with **--out-richness**, **--out-private** or **--out-tuples** uses that name literally and appends version 1.0's suffixes (`_summary`, `_fulldata`, `_deletedloci`) to it.

Results go to files; progress, warnings and errors go to standard error, so redirecting standard output captures results alone. **--quiet** silences the diagnostics. Progress bars are drawn only when standard error is a terminal.

5.1 Allelic richness and private allelic richness

One row per grouping per g : the grouping name, the g at which the statistics were computed, the number of loci used, and then the mean, variance and standard error across those loci (Equations 6–8). Groupings are separated by a blank line.

```
AMERICA 2 5 1.47429 0.0886378 0.133145
AMERICA 3 5 1.76929 0.264997 0.230216

EUROPE 2 5 1.68 0.0208889 0.0646357
```

5.2 Private allelic richness of tuples

The same layout, with the tuple's grouping names in place of the single name. With **--tuples-k** the tuple size k is appended to the file name, one file per k ; with **--tuples** the named tuples all go to one file.

5.3 Locus-specific output

--full-richness, **--full-private** and **--full-tuples** add a `_fulldata` file that carries, in addition to the summary columns, the statistic at every locus used. The first line names the columns, including one column per locus.

5.4 Filtered loci

When **--tolerance** is below 1, a `_deletedloci` file records how many loci were dropped, the threshold applied, and the name of each dropped locus.

5.5 Run summary

A `_summary` (or `.summary.txt`) file is written on every run. It echoes the parameters actually used — including values that were measured rather than declared, and `MAX_G` when it was resolved from the data — and records the elapsed time at which each phase finished.

5.6 Tab-separated output

`--tsv` writes the result files with a header row, tab separators, and `NA` in the value columns where a statistic is undefined at that g . Without it, the legacy format is used: no header, space separators, and rows that are undefined at a given g are omitted entirely. The legacy format remains the default so that existing scripts keep working, but `--tsv` is preferable for new work, because a consumer can distinguish “not computable here” from “not run”.

5.7 Exit status

0	success
2	usage error: unknown option, missing value, contradictory request
3	I/O error: a file could not be opened, read or written
4	data error: the file disagrees with itself or with the parameters

6 Examples

6.1 The distributed example data set

The data set in Figure 1 has five loci and three groupings. Running it with nothing but the data file and the grouping column,

```
adze --data small_data.stru --group-col 5 --out-prefix small
```

reports on standard error

```
Read 22 gene copies at 5 loci in 3 groupings (d:h:m:s 0:0:0:0.00)
MAX_G not given; using 3, the largest the 5 surviving loci support.
```

and writes richness results that stop at $g = 3$. Three is low for 22 gene copies, and the reason is missing data rather than sample size: the EUROPE grouping has six gene copies in total, but at locus D9S1825 three of those six are missing, leaving $N_j = 3$. D9S1825 is also missing 25% of the AMERICA grouping, and D14S1007 is missing 25% of EAST_ASIA.

Filtering those two loci buys a much larger standardized sample size:

```
adze --data small_data.stru --group-col 5 --out-prefix small24 \
--tolerance 0.24
```

```
2 loci have at least one grouping with more than 24% missing data.
MAX_G not given; using 6, the largest the 3 surviving loci support.
```

Now the analysis runs to $g = 6$, the limit set by EUROPE’s sample size alone, over the three surviving loci. Which of the two runs to prefer is a judgement about the data: the first uses all five loci but compares populations only at $g \leq 3$; the second compares them over a much wider range of g using three loci.

The parameter file distributed with the example, `small_paramfile.txt`, runs the same analysis in version 1.0’s style, declaring all four dimensions along with `MAX_G 8` and `TOLERANCE 0.2`:

```
adze small_paramfile.txt
```

Its results agree with the run above wherever the two overlap, and the private allelic richness files are identical. The richness files are not: declaring `MAX_G 8` reports `AMERICA` and `EAST_ASIA` at $g = 7$ and 8, which their own sample sizes reach, whereas the default stops at the largest g that *every* grouping supports. Private allelic richness stops at 6 in both, because no allele can be private at a g that some grouping cannot reach.

6.2 A larger data set

The next example uses the data of Rosenberg et al. (2002): 1056 individuals from 52 populations typed at 377 microsatellite loci, available at <https://rosenberglab.stanford.edu/diversity.html>. Because computing private alleles for all combinations at the population level would be prohibitive, the published analysis groups the individuals by region instead, which is column 5 of that file’s labels and reduces the number of groupings from 52 to 7.

The version 1.0 parameter file for this analysis (Figure 3 of the version 1.0 manual) declared `MAX_G 70`, `DATA_LINES 2112`, `LOCI 377`, `NON_DATA_ROWS 1`, `NON_DATA_COLS 5`, `GROUP_BY_COL 5`, `K_RANGE 2-6` and `TOLERANCE 1`. In version 2.0 the four dimensions are measured from the file, so the same run is

```
adze --data diversitydata.stru --group-col 5 --max-g 70 \
    --stat richness,private,tuples --tuples-k 2-6 \
    --out-prefix hgdp --threads 8
```

Figures 2–4 show the published analysis of these data: mean allelic richness against g , mean private allelic richness against g , and mean private allelic richness for combinations of six of the seven regions — that is, “missing” allelic richness, Equation 5. These figures are reproduced from the version 1.0 manual and were produced with version 1.0; version 2.0 computes the same values.

Africa carries both the highest allelic richness and, by a wide margin, the highest private allelic richness, while the Americas and Oceania have the highest missing allelic richness — the alleles absent there but present everywhere else.

6.3 Naming the tuples of interest

A full sweep over tuple sizes evaluates $\binom{J}{k}$ combinations for each k , which is 2^J tuples if every k is requested. When only a few groupings matter, list them in a file, one tuple per line:

```
# tuples.txt
AMERICA, EUROPE
EAST_ASIA
AMERICA, EUROPE, EAST_ASIA

adze --data small_data.stru --group-col 5 --tuples tuples.txt \
    --stat tuples --out-prefix regions
```

Each line is a comma- or space-separated list of grouping names as they appear in the data file; # starts a comment. This computes only the tuples named, at any mixture of sizes.

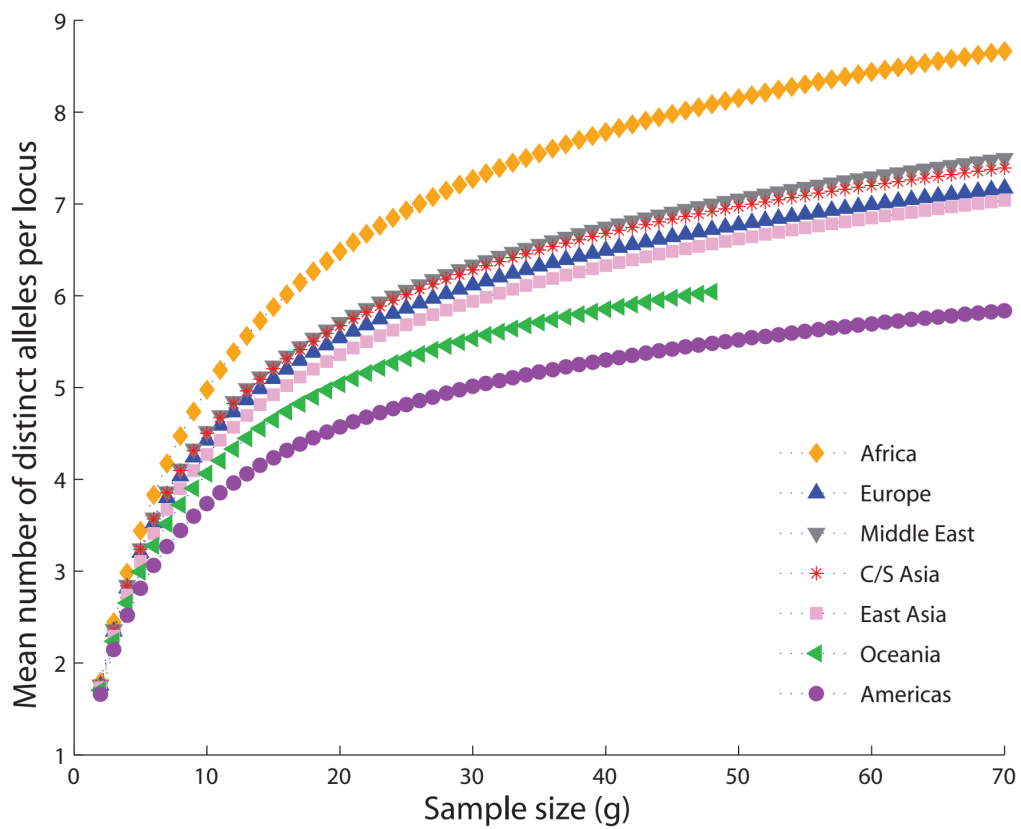


Figure 2: The mean number of distinct alleles per locus as a function of standardized sample size for seven major geographic regions.

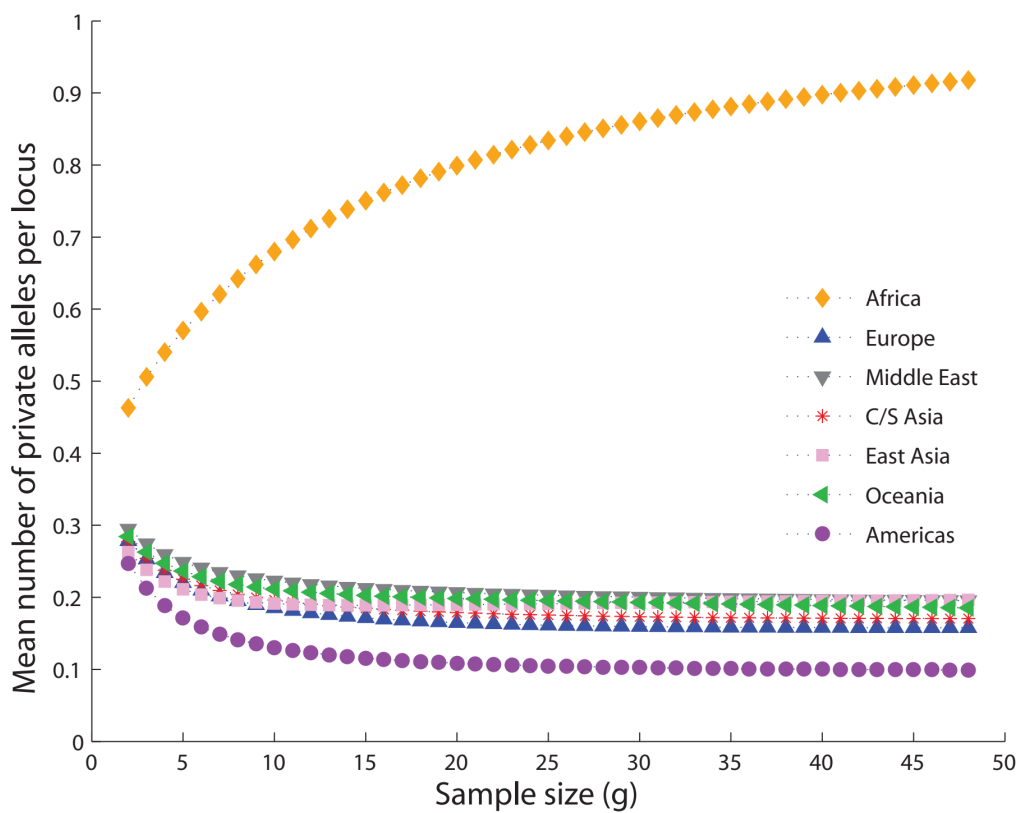


Figure 3: The mean number of private alleles per locus as a function of standardized sample size for seven major geographic regions.

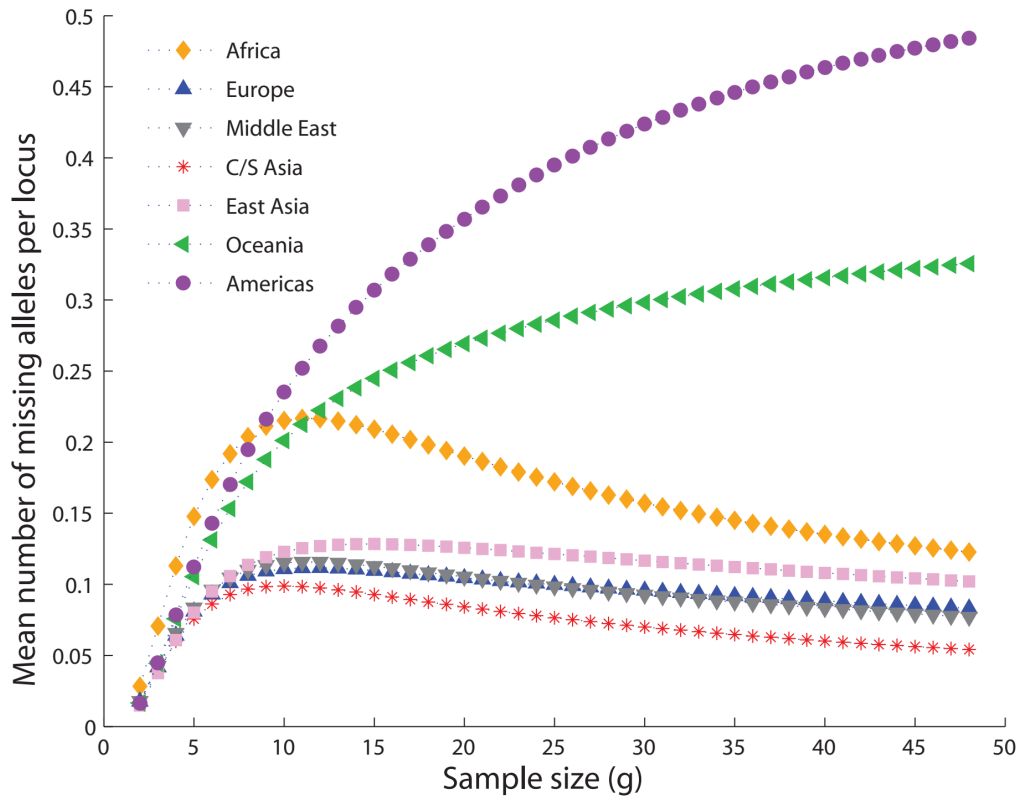


Figure 4: The mean number of missing alleles per locus for seven major geographic regions (private to combinations of six of the seven regions) as a function of standardized sample size.

7 Performance and threading

The per-locus loops of all three statistics can be spread over threads with `--threads` in an OpenMP build. Loci are processed independently and reduced in locus order, so the output does not depend on the thread count: the result files are bit-identical at any `--threads` setting.

Compressed input is decompressed as it streams, so a `.vcf.gz` costs more processor time to read than the same data uncompressed, and usually less wall-clock time on a busy filesystem, since there is far less to transfer.

Table 1 gives the effect of the version 2.0 work on the phases that dominate large runs, measured against a version 1.0 build on the same inputs. The costs that grew fastest in version 1.0 were locus filtering (quadratic in the number of loci) and the sweep over g (quadratic in `MAX_G`); both are now linear.

Phase	Condition	Version 1.0	Version 2.0
Locus filtering	8000 loci, 4623 dropped	5.97 s	<0.01 s
Richness + private	1000 loci, <code>MAX_G</code> 80	4.87 s	0.24 s
Tuples	12 groupings, 495 tuples	4.33 s	0.67 s
Reading	64 MB input	3.62 s	0.70 s
Peak memory	64 MB input	404 MiB	21 MiB
Statistics, 8 threads	2000 loci, <code>MAX_G</code> 150	—	4.0×

Table 1: Measured on an 8-core machine, best of two runs. Version 1.0 held every genotype in memory, so its footprint grew with sample size; version 2.0 keeps allele counts only, and its footprint is flat in the number of gene copies.

8 Changes from version 1.0

8.1 Compatibility

Parameter files written for version 1.0 work unchanged, as do the short flags. Explicitly named output files keep version 1.0’s literal suffixes, and the default result format is unchanged. `SKIP_CHK` and `--no-check` are still accepted but do nothing, because validation now costs nothing: it happens during the single read pass.

8.2 Interface

- Options can be given without a parameter file, and `--help` and `--version` work with no arguments at all. A bare invocation prints usage; version 1.0 wrote a template parameter file into the working directory.
- The parameter-file parser reads a keyword and the rest of the line. Version 1.0 searched each line for any keyword as a substring, so a value that contained a keyword — `R.OUT out_LOCI_richness` — was rejected, and values could not contain spaces.
- Dimensions are measured rather than declared (section 3.3).
- Diagnostics and progress go to standard error; exit status distinguishes usage, I/O and data errors.

- VCF input, plain or `gzip/bgzip` compressed, via `--format` and `--samples` (section 3.4). Version 1.0 read only the uncompressed STRUCTURE layout.
- New: `--out-prefix`, `--stat`, `--pops`, `--exclude-pops`, `--tuples`, `--tsv`, `--dry-run`, `--threads`, `--quiet`.

8.3 Behaviour

Estimates are unchanged. The following differ, in each case because version 1.0 gave an answer that could not be used:

1. A locus with no observed allele in any grouping is reported as undefined, with a warning. Version 1.0 stopped with a segmentation fault.
2. Filtering that removes every locus is an error. Version 1.0 continued and printed `nan` rows over zero loci.
3. A data row whose column count is wrong is always an error. With checking disabled, version 1.0 would misread it silently.
4. The variance of a single locus is reported as `nan` rather than the result of dividing by zero.
5. Reported times are elapsed wall-clock times with two decimals. Version 1.0 reported processor time truncated to whole seconds, which is why the example run in the version 1.0 manual reports 0:0:0:0 throughout.

8.4 Verification

Every change was checked with a differential test that runs version 2.0 and a version 1.0 build over the same inputs and compares every output file byte for byte: nine synthetic data sets — spanning allele-count range, missingness, a single locus, a single grouping, unequal sample sizes, haploid data, and a locus with no observed alleles — crossed with nine parameter settings, plus the distributed example. All 69 comparable cases are byte-identical; the remaining cases are those in which version 1.0 crashes or reports statistics over zero loci, and so has no output to compare. The test is in `test/regress.py` and needs a version 1.0 binary to compare against.

VCF input has no version 1.0 counterpart, so it is checked differently: `test/formats.py` writes one set of genotypes as STRUCTURE, as VCF, and as compressed VCF, and requires the three runs to agree byte-for-byte. Because both readers order allele slots by the same rule, the comparison is exact rather than approximate — any difference at all is a real one, not a change in the order of a floating-point sum.

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